Input Form (IF)

Score: 2/5 — Significant gaps | Confidence: high

***Benchmark:** WXIMPACTBENCH* | ***Context:** Argentine Government Disaster Reporting Reliability — Policy Staff*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark is text-only [Q19], which matches the deployment's text-based channels. However, the form characteristics diverge substantially: benchmark inputs are long-form (avg 2,987 tokens) or chunked (avg 781 tokens) [Q109], formally structured historical journalism with 'outdated terminology, spelling variations, and evolving writing conventions' [Q14]. The Argentine deployment includes social media posts (often <280 chars), WhatsApp messages, hashtags, emoji, lunfardo slang, and informal abbreviation — none of which is represented. Critically, the paper itself notes that formal historical narrative style is *easier* for LLMs to classify [Q83, Q85], meaning benchmark performance estimates likely overstate model capability on noisier informal inputs. Web research confirms most LLMs face specific challenges with flood-related social media even in English [WEB-17]. The benchmark is English-only with Latin script; Rioplatense Spanish, while also Latin-script, introduces voseo, yeísmo rehilado lexicon, and Italian-influenced vocabulary not exercised. Empirically, the dataset contains heavily OCR-corrupted financial table chunks [DATASET-D23] which are not analogous to Argentine social media noise but do confirm noise tolerance is partially exercised. Input form is a MODERATE-priority dimension per elicitation; score reflects significant register and length mismatch.

ID	Evidence
Q14	“Historical newspapers often employed more descriptive and elaborate narratives compared to modern reporting styles (Bingham, 2010). These narratives frequently included outdated terminology, spelling variations, and evolving writing conventions (Campbell, 2013).” (p.2)
Q19	“The construction of the dataset aims to obtain high-quality text samples. The pipeline overview is presented in Figure 2, which consists of four stages: data collection, post-OCR correction, topic-aware article selection, and manual label annotation.” (p.3)
Q83	“The reason might be the structured and formal narrative style used to report disruptive weather events in historical periods, which more explicitly highlights cause-and-effect relationships.” (p.8)
Q85	“Though the modern text might dominate within the pre-trained corpus, the language patterns used in historical narrative styles are easier for language models to identify, and thus perform better on classifying disruptive weather impacts.” (p.8)
Q109	“The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings.” (p.15)
D23	-2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25..." / "Escenvt 115780 298 J80 289 19 Caiunoil 8000 J9 J9 29... — OCR-corrupted stock market tables; repeated for same article id across multiple rows
WEB-17	Most LLMs face specific challenges processing flood-related social media data even in English (HumAID-based six-LLM analysis) (https://www.semanticscholar.org/paper/HumAID:-Human-Annotated-Disaster-Incidents-Data-Alam-Qazi/462cc2046ef4d48d84481 3b66d8a1ed6dfda3bc0)

Strengths

- Text modality matches deployment's text channels [Q19]
- Mixed-context chunking (avg 781 tokens) [Q109] is closer to press release length than full historical articles
- Two linguistic registers (historical and modern) exercise some register robustness [DATASET-D14, D18]
- OCR-noisy chunks in dataset [DATASET-D23] provide some test of noise tolerance

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Q109	“The average number of tokens per article is 2987.4 in long-context settings and 781.3 in mixed-context settings.” (p.15)
D14	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed... an unprecedented snowfalls and heavy winds. — 1894 blizzard; infrastructure (transit) disruption labeled; North American geography
D18	more than 100,000 homes damaged and crops on about 175,000 hectares of farmland destroyed... death toll in Chongqing in China's southwest rose to 42 people and 12 missing from torrential downpours — 2007 global flood roundup (England, China, Pakistan) — non-Canadian events used as positive examples
D23	-2 City res y 5000 5 5 5 Aiexavn 8000 50 50 50 Ckearcdnf 835 825 25..." / "Escenvt 115780 298 J80 289 19 Caiunoil 8000 J9 J9 29... — OCR-corrupted stock market tables; repeated for same article id across multiple rows

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Significant signal mismatch: benchmark long-context avg 2,987 tokens, mixed-context avg 781 tokens [Q109]; deployment includes tweet-length (<280 chars), WhatsApp messages, and government press bulletins. Register mismatch: formal English journalism vs. Rioplatense Spanish informal social media with lunfardo, emoji, hashtags.

IF-2: Argentine infrastructure supports text capture: 90.6% national internet penetration, 98.4% mobile broadband [WEB-5], Buenos Aires fixed broadband 41–101 Mbps [WEB-7]. However, rural disaster-zone producers have lower connectivity (three provinces <20 Mbps) [WEB-7, WEB-8], potentially yielding compressed/delayed content not represented in benchmark.

IF-3: Domain-specific form differences: deployment requires handling Twitter/X short-form, WhatsApp messages, emoji, hashtags, code-switching English/Spanish, OCR-absent native digital text — none in benchmark [Q19, Q109].

IF-4: External validity harm is significant: paper explicitly notes formal historical style is easier for LLMs [Q83, Q85], and HumAID-based research shows LLMs struggle with flood social media even in English [WEB-17]. Performance metrics on this benchmark likely overstate Argentine deployment capability on informal channels.

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WEB-5	98.4% Argentine mobile broadband indicates short-form mobile content dominant in deployment (https://datareportal.com/reports/digital-2026-argentina)
WEB-7	Rural-urban digital divide means disaster-zone content carries different noise characteristics than urban Buenos Aires (https://freedomhouse.org/country/argentina/freedom-net/2024)
WEB-8	https://ts2.tech/en/state-of-internet-access-in-argentina-fiber-5g-and-satellite-in-2025/
WEB-17	Most LLMs face specific challenges processing flood-related social media data even in English (HumAID-based six-LLM analysis) (https://www.semanticscholar.org/paper/HumAID:-Human-Annotated-Disaster-Incidents-Data-Alam-Qazi/462cc2046ef4d48d844813b66d8a1ed6dfda3bc0)

Information Gaps

- Empirical token-length distribution of Argentine disaster reporting across formal press, press releases, and social media
- Quantitative LLM performance gap between formal long-form and informal short-form Spanish disaster content

Requires Expert Verification

- Whether the mixed-context (~781 token) chunks adequately approximate Argentine government press bulletin length

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Remediation

Gap 1: Long-form formal register only; deployment includes short-form informal Spanish social media [Q83, Q109, WEB-17]

Recommendation: Run a parallel evaluation on Spanish-translated tweet-length disaster content and on Argentine government press bulletins (~100–300 tokens) to quantify performance degradation relative to formal long-form benchmarks. Calibrate downstream credibility scoring to discount benchmark performance on informal channels by the measured gap.

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